SOFTWARE DESIGN SPECIFICATION

Kanboard Project Management System

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Submitted To: Prof. Swathi J.N.

Signatures

Date	Revision	Approved By
2026.02.23	1.0	Prof. Swathi J.N.

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Change History

Revision	Date	Description
1.0	27.02.2026	Initial Revision — SDS for Kanboard Project Management System

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Preface

This document presents the Software Design Specification (SDS) for the Kanboard Project Management System — an academic project involving the reverse and forward engineering of Kanboard, an open-source Kanban-based project management tool. The project is submitted in partial fulfilment of the Software Engineering course requirements at VIT University.

The major sections of this document address the system decomposition by module, concurrent process, and data entity, as well as inter-module dependencies, interface descriptions, and detailed design of all identified components.

Section 2, Decomposition Description, presents the high-level system architecture of the Kanboard system, including its MVC module breakdown, concurrent processes, and data entities. The three-tier web-based architecture is explained with appropriate context.

Section 4, Interface Description, describes the user interfaces and process logic for each module of the Kanboard system. Module interfaces for all core use cases are provided with process-level descriptions.

Section 5, Detailed Design, extends the discussion from Section 2 and provides class-level design details for each system module, including UML diagram placeholders for implementation in Modelio.

1 Introduction

1.1 Purpose

The purpose of this Software Design Specification is to describe the detailed design of the Kanboard Project Management System. The system is a web-based, three-tier Kanban project management tool. This document provides comprehensive module decomposition, dependency analysis, interface descriptions, and detailed design artifacts required for implementation and academic evaluation.

The design specification encompasses user interface design, object-oriented class design, process descriptions, and data design for all identified system modules.

1.2 Scope

This document covers the design of the Kanboard Project Management System as identified through reverse engineering of the original open-source Kanboard tool and extended through forward engineering to include proposed enhancements.

Within scope: user interface design, MVC architectural decomposition, object-oriented class design, process flow design, database entity design, and inter-module dependency mapping. Out of scope: deployment infrastructure, third-party API internals, and mobile client design.

1.3 Definitions and Acronyms

Definition / Acronym	Description
SDS	Software Design Specification
SRS	Software Requirements Specification
MVC	Model-View-Controller Architectural Pattern
UML	Unified Modelling Language
CRUD	Create, Read, Update, Delete
UI	User Interface
DB	Database
PHP	Hypertext Preprocessor (server-side scripting language)
HTTP	Hypertext Transfer Protocol
RBAC	Role-Based Access Control
ER	Entity Relationship
DFD	Data Flow Diagram
API	Application Programming Interface

1.4 References

Reference	Description
Software Requirements Specification	SRS document for the Kanboard Project Management System (Kushagra Dixit, 2026)
Project Abstract	Project Idea Submission — Reverse and Forward Engineering of Kanboard (23BKT0121)
Kanboard Official Documentation	https://docs.kanboard.org/
Kanboard GitHub Repository	https://github.com/kanboard/kanboard
IEEE 830	IEEE Recommended Practice for Software Requirements Specifications
SDS Sample Template	Seminole Software — Electronic Stamp Project SDS (Version 1.2, 2003)

2 Decomposition Description

2.1 Module Decomposition

The Kanboard Project Management System has been decomposed into the following primary software modules, following the Model-View-Controller (MVC) architectural pattern implemented across a three-tier web architecture:

- User Authentication Module: Manages user registration, login, session handling, and role-based access control (RBAC). Interfaces with the User Model and Session Manager.
- Project Management Module: Handles the creation, configuration, and deletion of projects. Manages project-level user assignments and permissions.
- Board Management Module: Manages the creation of Kanban boards within projects. Handles column definition, swimlane configuration, and board rendering logic.
- Task Management Module: Core module responsible for creating, reading, updating, and deleting tasks. Manages task attributes including title, description, due date, assignee, priority, and colour coding.
- Task Assignment and Workflow Module: Manages the assignment of tasks to team members and the movement of tasks across board columns (workflow progression).
- Comment and Collaboration Module: Allows users to add comments to tasks, supports @mentions, and maintains a task activity history log.
- Admin Management Module: Provides administrative controls for user management, system configuration, role assignment, and global settings.

- Notification Module (Enhancement): Manages automated notifications triggered by task events such as assignment, due-date proximity, and status changes.
- Analytics Dashboard Module (Enhancement): Aggregates task data to present metrics including cycle time, throughput, work-in-progress counts, and burndown visualisations.

2.2 Concurrent Process Decomposition

The Kanboard Project Management System involves two primary concurrent processes:

Web Client Process: Executes within the user's browser. Handles user interactions, renders views using HTML/CSS/JavaScript, and communicates with the server via HTTP requests. Drag-and-drop task movement is handled client-side using HTML5 APIs.

Web Server Process: Executes on the PHP application server. Receives HTTP requests from the web client, invokes appropriate Controller classes, coordinates with Model classes for business logic and database operations, and returns rendered views to the client.

Additionally, a Background Job Process handles scheduled automation tasks such as due-date notification generation and automation rule execution. This process runs independently of user interactions via server-side cron jobs.

These processes run concurrently. The client process and server process exchange data exclusively through HTTP requests and responses. The background process interacts only with the database and notification subsystem.

2.3 Data Decomposition

The following are the primary data entities managed by the Kanboard system:

User Entity: Stores user account data including username, hashed password, email address, role designation (Admin, Project Manager, Team Member), and account status.

Project Entity: Stores project configuration including project name, description, owner, creation date, and status. A project acts as the top-level container for boards and tasks.

Board Entity: Stores board configuration including the associated project, board name, column definitions, and swimlane settings. Boards represent the visual Kanban workspace.

Column Entity: Represents a workflow stage within a board (e.g., Backlog, In Progress, Done). Stores column name, position order, task limit (WIP limit), and parent board reference.

Task Entity: The central data entity. Stores task title, description, assignee, creator, project, column, swimlane, priority, colour, due date, creation date, completion date, and status.

Comment Entity: Stores user comments associated with specific tasks. Includes comment text, author, timestamp, and parent task reference.

Automation Rule Entity (Enhancement): Stores automation rule definitions including trigger conditions, associated actions, and execution history.

3 Dependency Description

3.1 Inter-module Dependencies

3.1.1 Independent Modules

The following modules are independent and do not rely on any other system module for initiation or data provision:

- User Authentication Module: Initiates independently on application startup. All other modules depend on this module for session validation.
- Admin Management Module: Operates independently, providing system-wide configuration and user administration functions.

3.1.2 Dependent Modules

The following modules are dependent on one or more other modules for their operation:

- Project Management Module: Depends on the User Authentication Module for user identity and role verification before permitting project creation or modification.
- Board Management Module: Depends on the Project Management Module. A board cannot exist without an associated project. Also depends on the User Authentication Module for access control.

- Task Management Module: Depends on the Board Management Module for board and column context. Task creation requires a valid board and column reference. Depends on the User Authentication Module for creator and assignee resolution.
- Task Assignment and Workflow Module: Depends on the Task Management Module and User Authentication Module. Task movement and assignment require valid task and user references.
- Comment and Collaboration Module: Depends on the Task Management Module for task context and the User Authentication Module for author identification.
- Notification Module: Depends on the Task Management Module and User Authentication Module to resolve notification triggers and recipient details.
- Analytics Dashboard Module: Depends on the Task Management Module, Board Management Module, and Project Management Module for data aggregation.

3.2 Inter-process Dependencies

The Web Client Process depends on the Web Server Process for all data retrieval and persistence operations. All client-side state changes that require persistence must be communicated to the server process via HTTP requests.

The Background Job Process depends on the database layer for reading automation rule definitions and task state. It also depends on the Notification Module for dispatching generated notifications. The background process does not depend on the Web Client or Web Server processes for its operation.

3.3 Data Dependencies

The following dependencies exist at the data level:

- Task data depends on Board and Column data — a task must be associated with a valid column within a board.
- Board data depends on Project data — a board must belong to a project.
- Comment data depends on Task data — comments are attached to tasks.
- Column data depends on Board data — columns are children of boards.
- Automation Rule data depends on Task and Project data — rules are scoped to projects and act on tasks.

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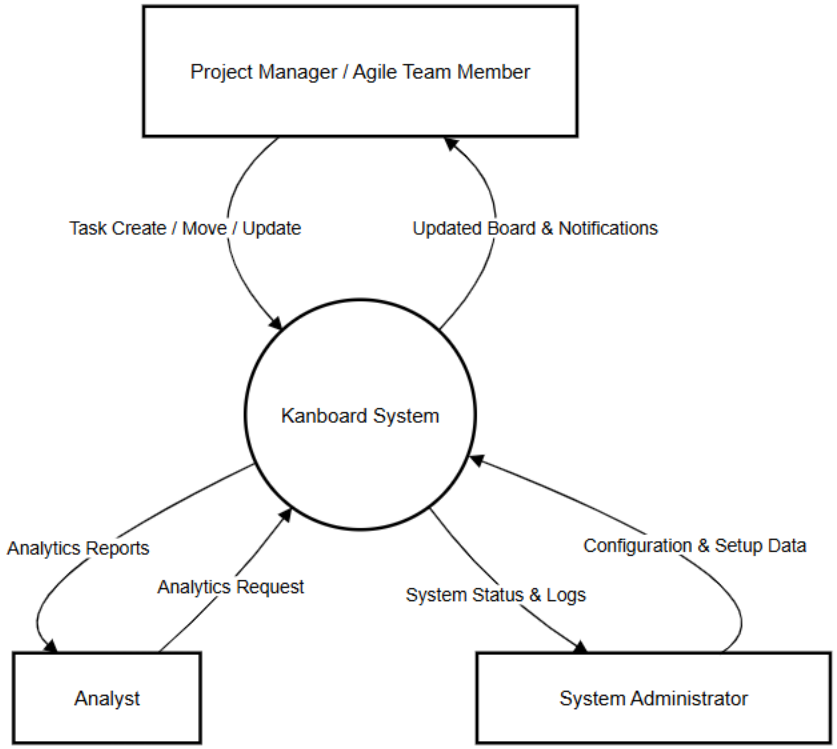

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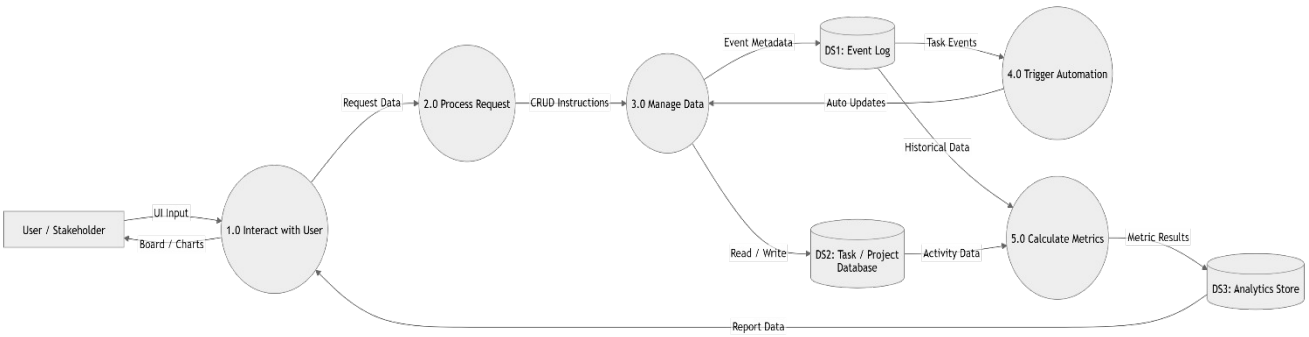


4 Interface Description

4.1 Module Interface

4.1.1 User Authentication Module Description

4.1.1.2 Description

The User Authentication Module provides the entry point for all system interactions. The Login UI presents a username and password form. Upon submission, credentials are validated against the database. On successful authentication, a session is created and the user is redirected to the project dashboard. Registration is accessible to new users and captures username, email address, password, and role selection. Role assignment may require administrator approval depending on system configuration.

4.1.2 Project Management Module Description

4.1.2.1 Description

The Project Management Module allows authenticated users to create, view, and manage projects. The Project Dashboard lists all projects accessible to the user, grouped by role. The Create Project interface captures project name, description, and owner. Project Managers and Admins may invite team members and configure project-level permissions from within this module.

4.1.3 Board Management Module Description

4.1.3.1 Description

The Board Management Module renders the primary Kanban workspace. Boards display columns representing workflow stages, with task cards positioned within appropriate columns. Users can create boards, add and reorder columns, and configure Work-In-Progress (WIP) limits per column. The board view is rendered server-side and enhanced client-side with drag-and-drop functionality using the HTML5 Drag-and-Drop API.

4.1.4 Task Management Module Description

4.1.4.1 Description

The Task Management Module provides the interface for creating and managing tasks. The Create Task form collects task title, description, assignee, due date, priority level, and target column. The Task Detail View displays full task information alongside the comment thread and activity history. Tasks may be edited, closed, or deleted from this view subject to role permissions.

4.1.5 Task Assignment and Workflow Module Description

4.1.5.1 Description

The Task Assignment and Workflow Module enables task assignment to specific users and movement of tasks across board columns. Assignment is performed via a user selector within the task detail view. Workflow progression is achieved through drag-and-drop on the board interface; alternatively, users may update the column field within the task detail view. The module enforces WIP limits and records movement events in the activity log.

4.1.6 Comment and Collaboration Module Description

4.1.6.1 Description

The Comment and Collaboration Module provides a threaded comment interface within the Task Detail View. Authenticated users may add text comments using Markdown syntax. Comments display the author name, timestamp, and formatted content. The activity history section below the comment thread records all task events including creation, assignment changes, column movements, and due date modifications.

4.1.7 Admin Management Module Description

4.1.7.1 Description

The Admin Management Module provides a dedicated administrative interface accessible only to users with the Admin role. Functions include user account management (create, deactivate, assign roles), global system configuration (authentication settings, notification preferences), and audit log review. The admin panel is accessible via a separate navigational route with server-side role enforcement.

4.2 Process Interface

4.2.1 User Authentication Process Description

The primary objective of this module is to authenticate user identity and establish a secure session. The system validates submitted credentials against hashed password records in the database using PHP password_verify(). On successful authentication, a session token is generated and stored server-side. Role information is loaded into the session and used by all subsequent modules for access control decisions. The session persists for a configurable duration and is invalidated on explicit logout.

4.2.2 Project Management Process Description

This module processes project creation and configuration requests. On project creation, the system validates the project name for uniqueness within the user scope, creates the project database record, assigns the creating user as Project Manager, and initialises a default board structure. Project retrieval queries are filtered by user role — Admins see all projects; Project Managers and Team Members see only assigned projects.

4.2.3 Board Management Process Description

The Board Management Process renders the Kanban board workspace. On board load, the controller retrieves the board record, all associated columns (ordered by position), and all tasks within those columns from the database. Data is passed to the view layer for server-side rendering. The view generates HTML column structures with embedded task cards. Client-side JavaScript attaches drag-and-drop event handlers to task cards and column drop zones.

4.2.4 Task Management Process Description

This process manages the complete lifecycle of a task. Task creation validates all required fields and creates a task record in the database, triggering a task_created event within the event system. Task updates follow an optimistic update pattern. Task closure records completion date and transitions the task to the configured completion column. Task deletion is soft-deleted by default, preserving audit history. All operations are gated by role permission checks.

4.2.5 Task Assignment and Workflow Process Description

The workflow process follows this sequence: when a user drags a task card to a new column (or submits a column update), an HTTP request is dispatched to the Task Controller with the task ID and target column ID. The controller validates that the user has permission to move tasks and that the target column WIP limit is not exceeded. On validation success, the task record is updated, a task_moved event is dispatched, and a success response is returned to the client for UI update.

4.2.6 Comment and Collaboration Process Description

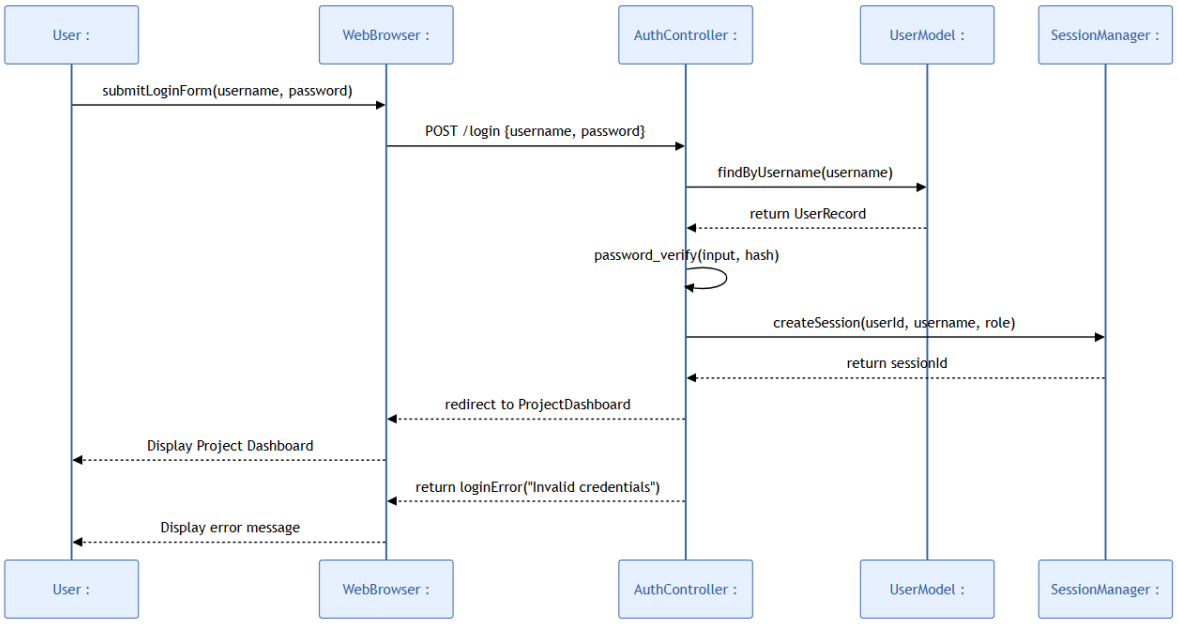
The comment process receives a comment submission containing the task ID, author user ID, and comment text. The controller validates the task existence and user authentication. The comment is stored in the comments table with a UTC timestamp. A task_commented event is dispatched to the notification subsystem. The updated comment thread is returned to the client and rendered in the Task Detail View. Markdown content is sanitised and rendered server-side.

5 Detailed Design

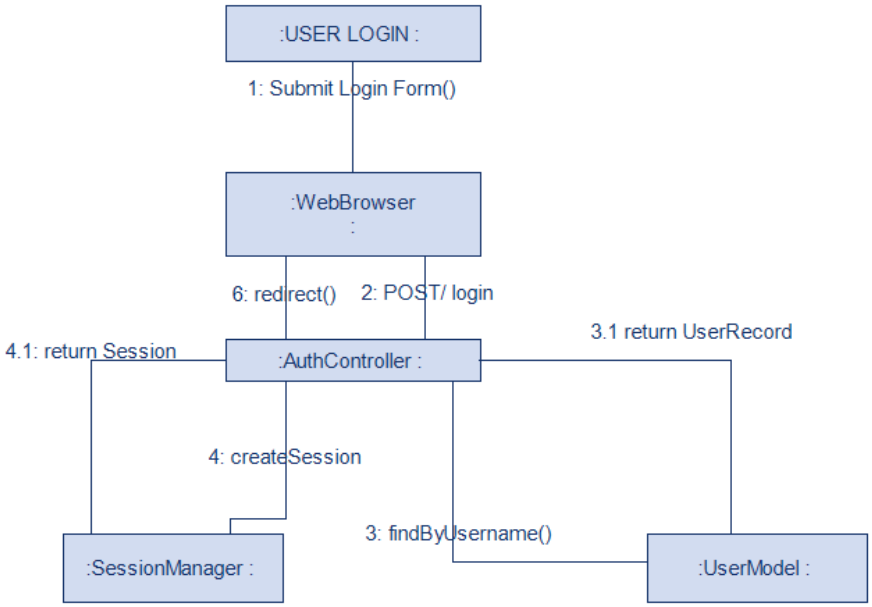
5.1 Module Detailed Design

5.1.1 Authentication Module Detailed Design

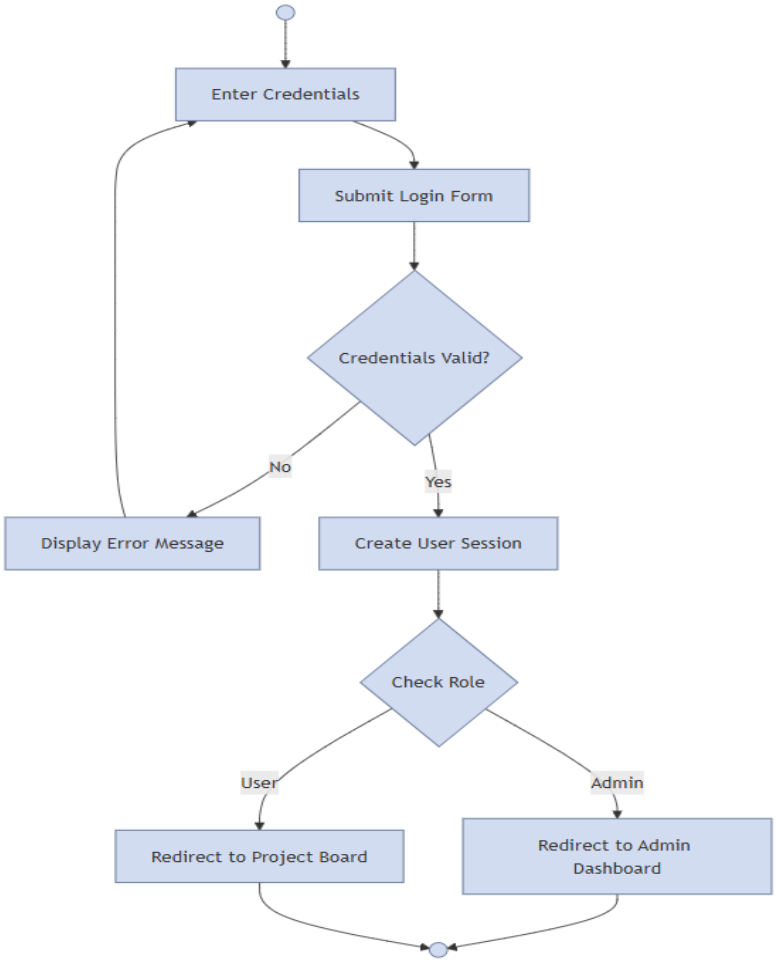
5.1.1.1 Sequence Diagram



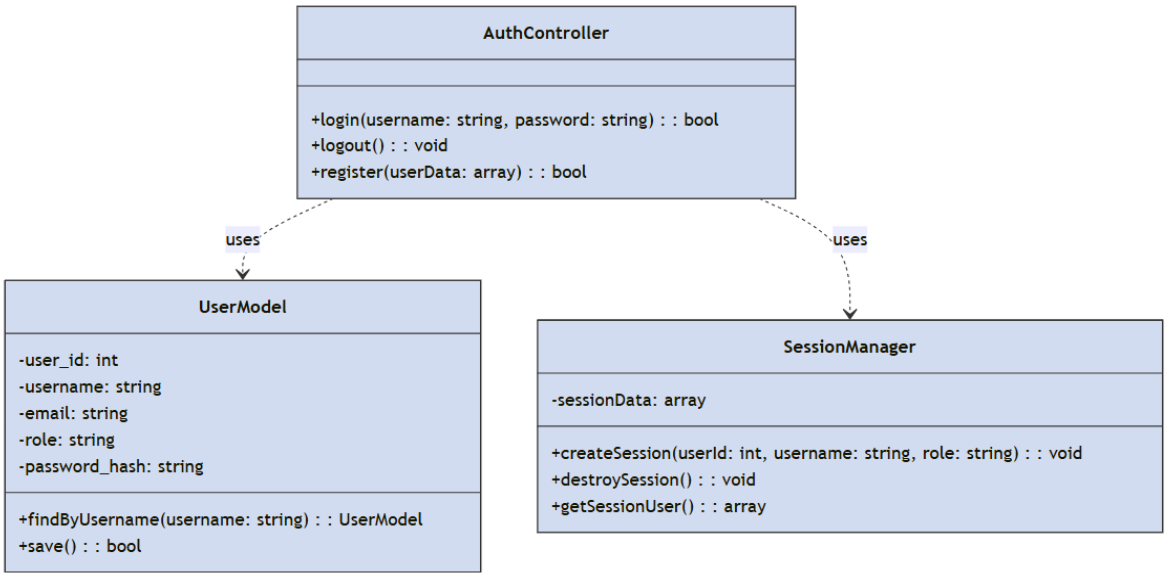
5.1.1.2 Collaboration (Communication) Diagram



5.1.1.3 Activity Diagram

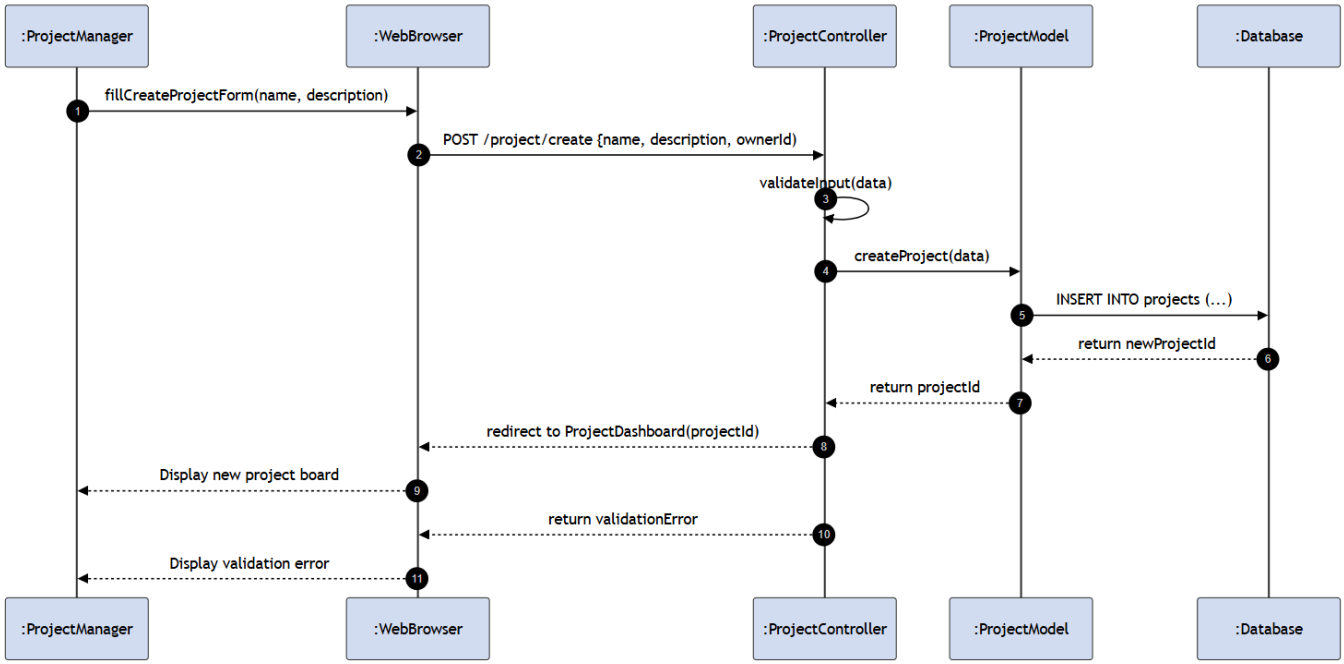


5.1.1.4 Class Diagram (module-level detailed class view)

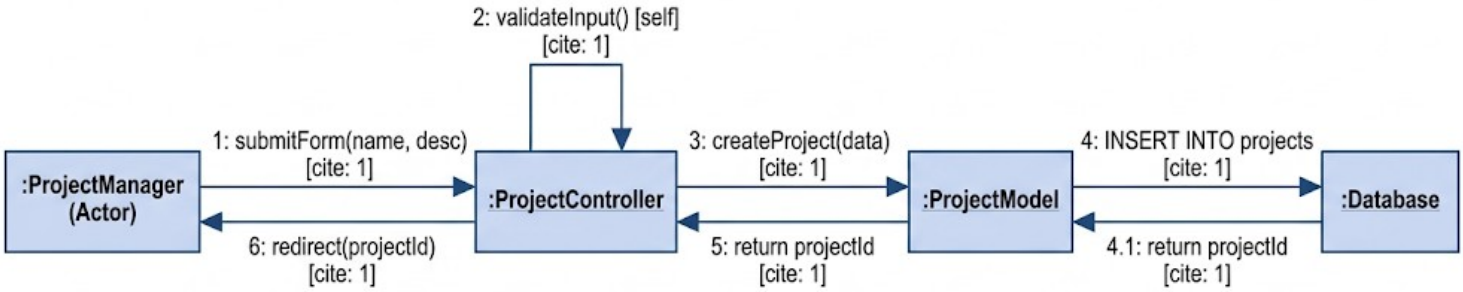


5.1.2 Project Management Module Detailed Design

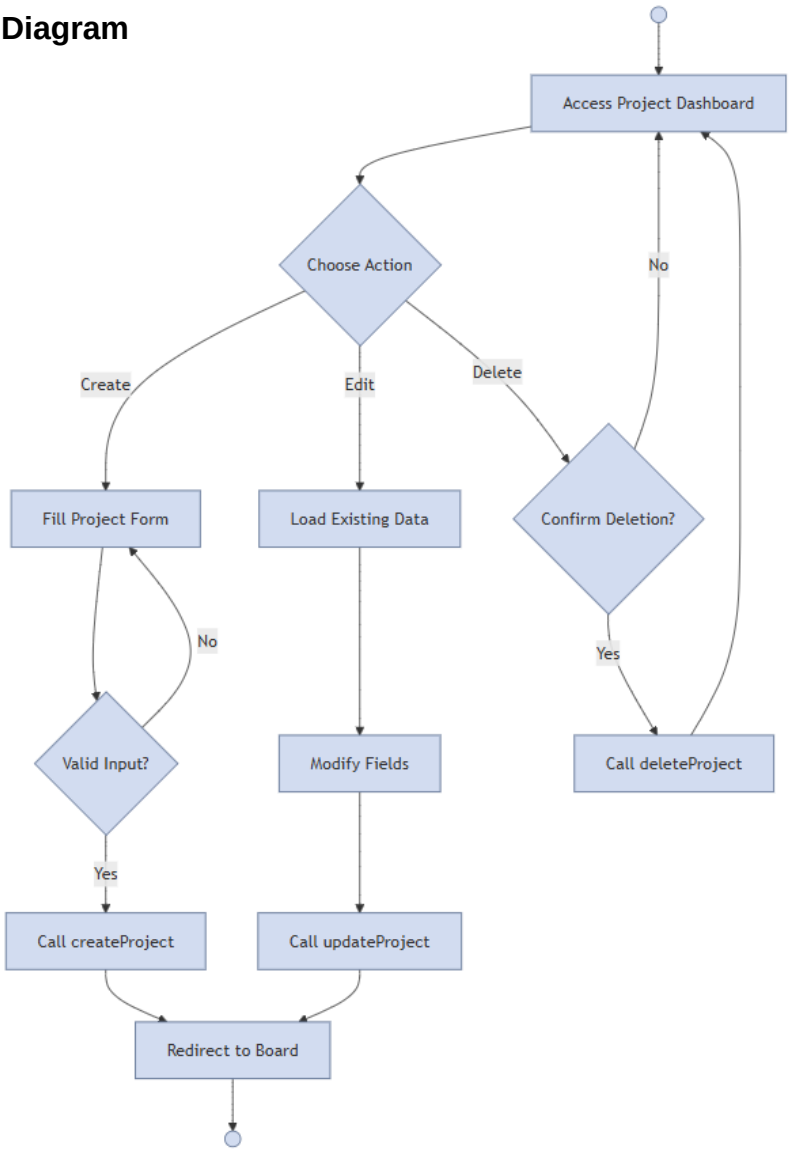
5.1.2.1 Sequence Diagram



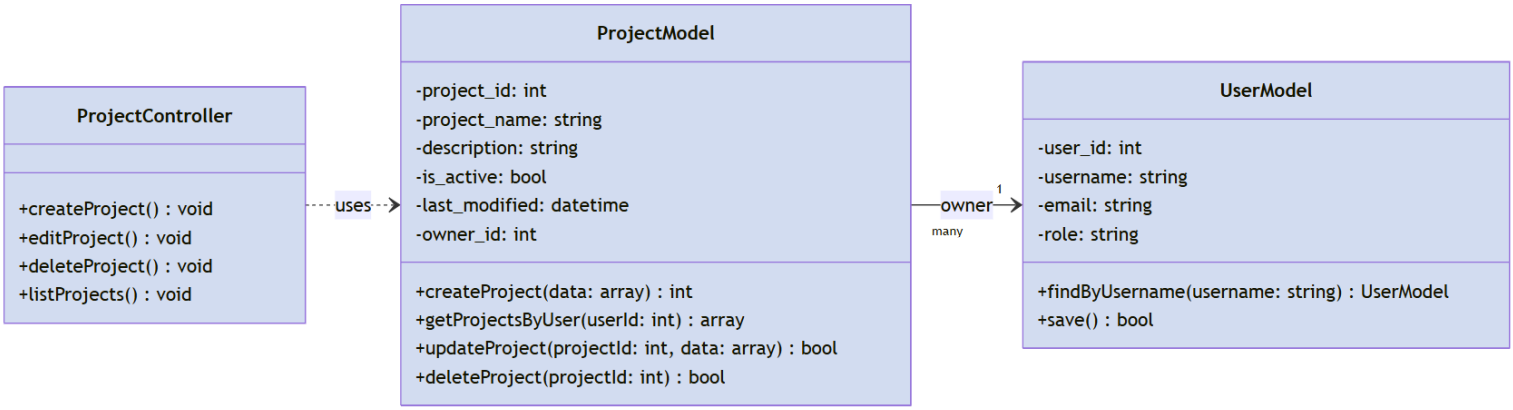
5.1.2.2 Collaboration (Communication) Diagram



5.1.2.3 Activity Diagram

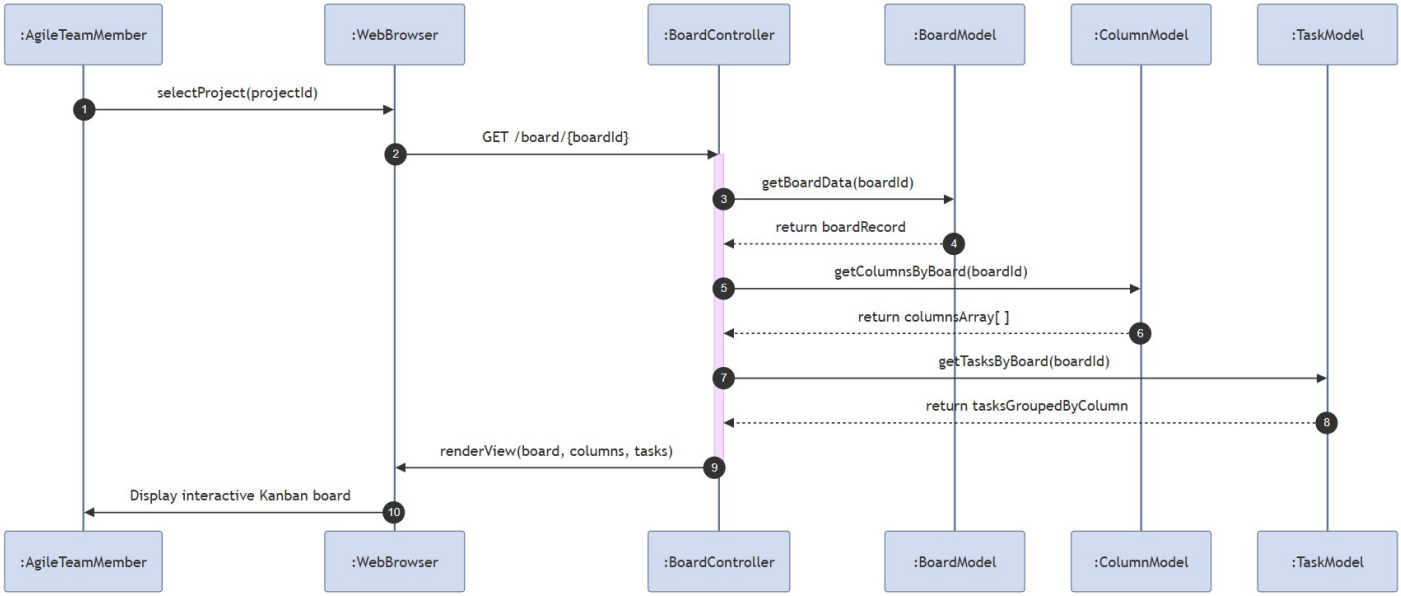


5.1.2.4 Class Diagram (module-level detailed class view)

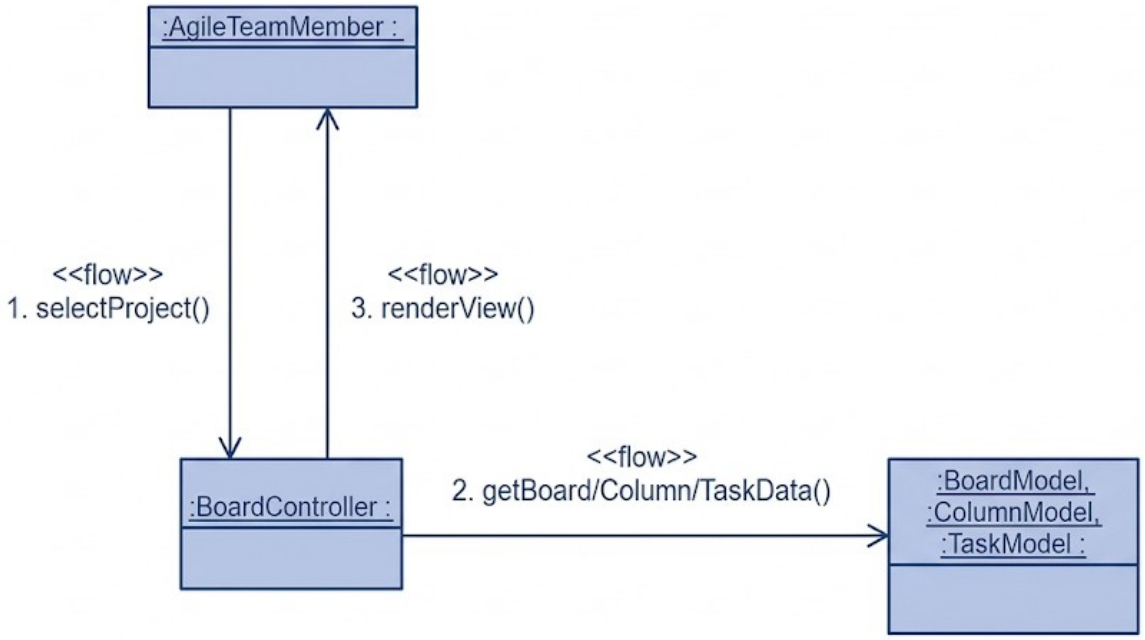


5.1.3 Board Management Module Detailed Design

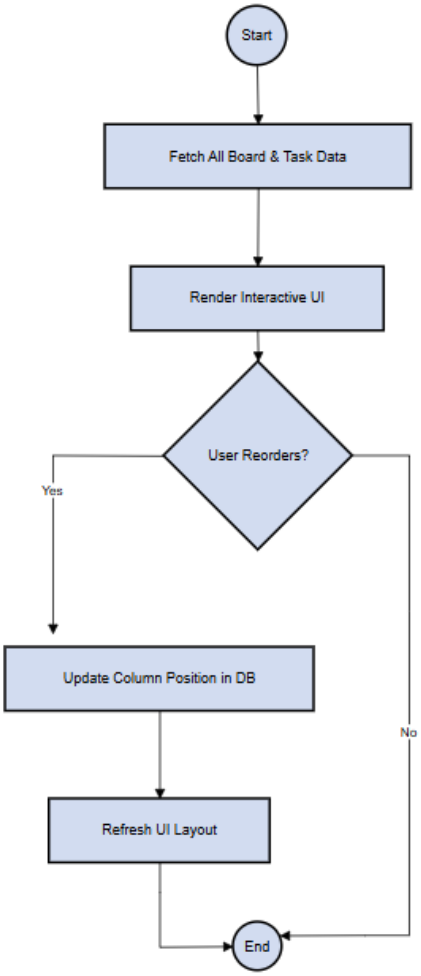
5.1.3.1 Sequence Diagram



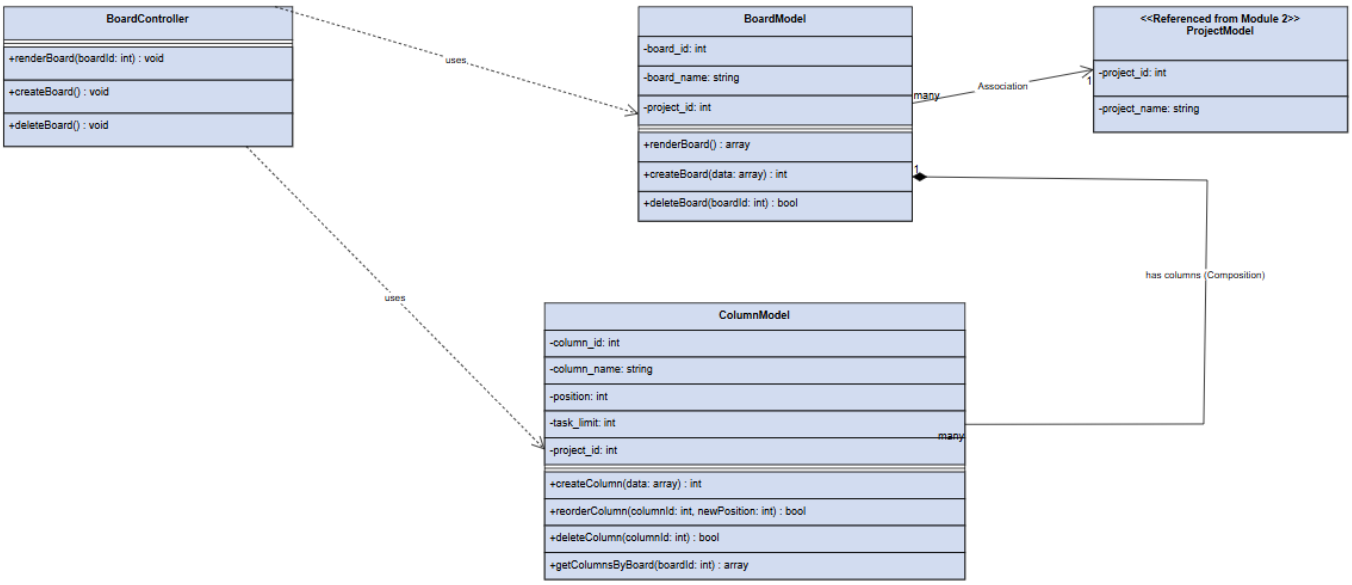
5.1.3.2 Collaboration (Communication) Diagram



5.1.3.3 Activity Diagram

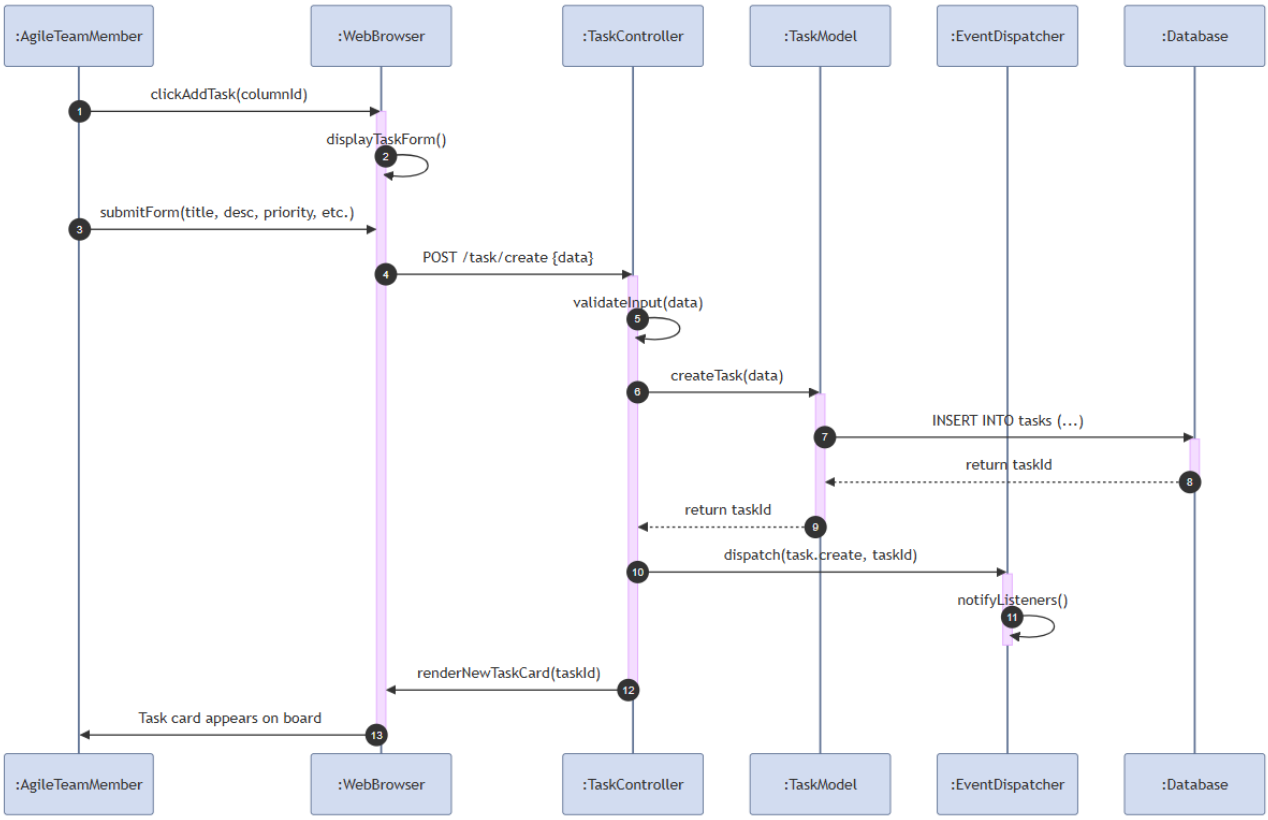


5.1.3.4 Class Diagram (module-level detailed class view)



5.1.4 Task Management Module Detailed Design

5.1.4.1 Sequence Diagram



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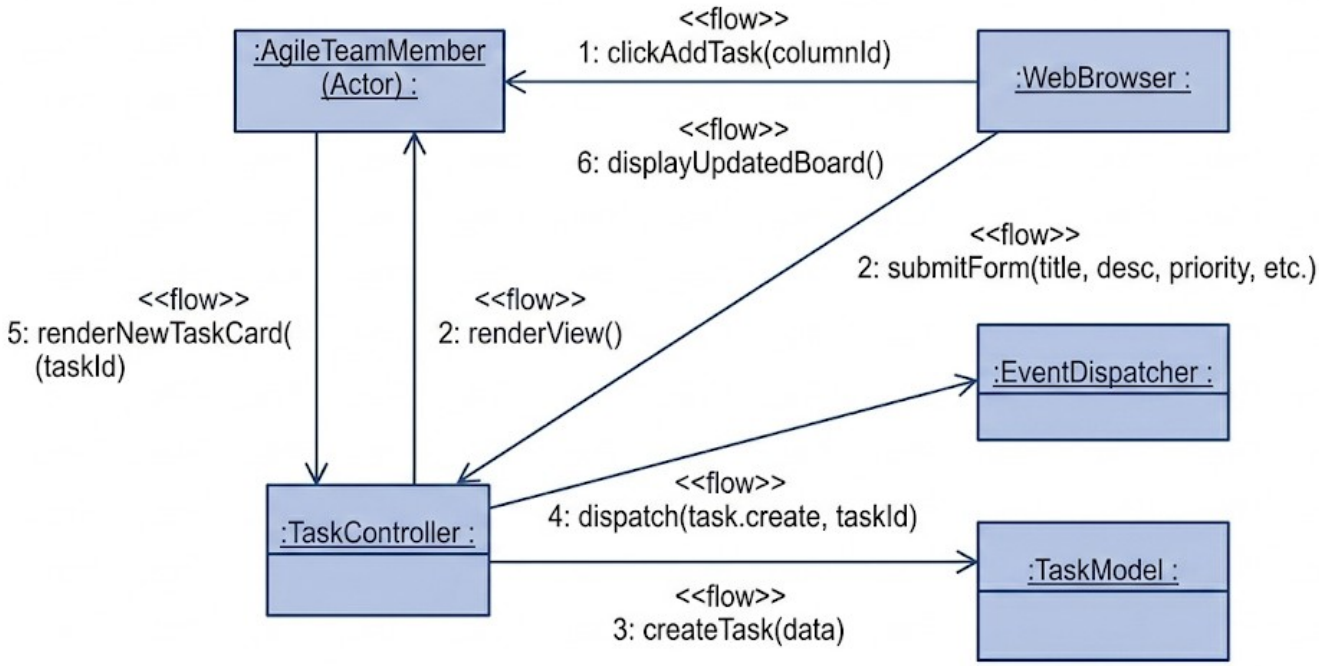
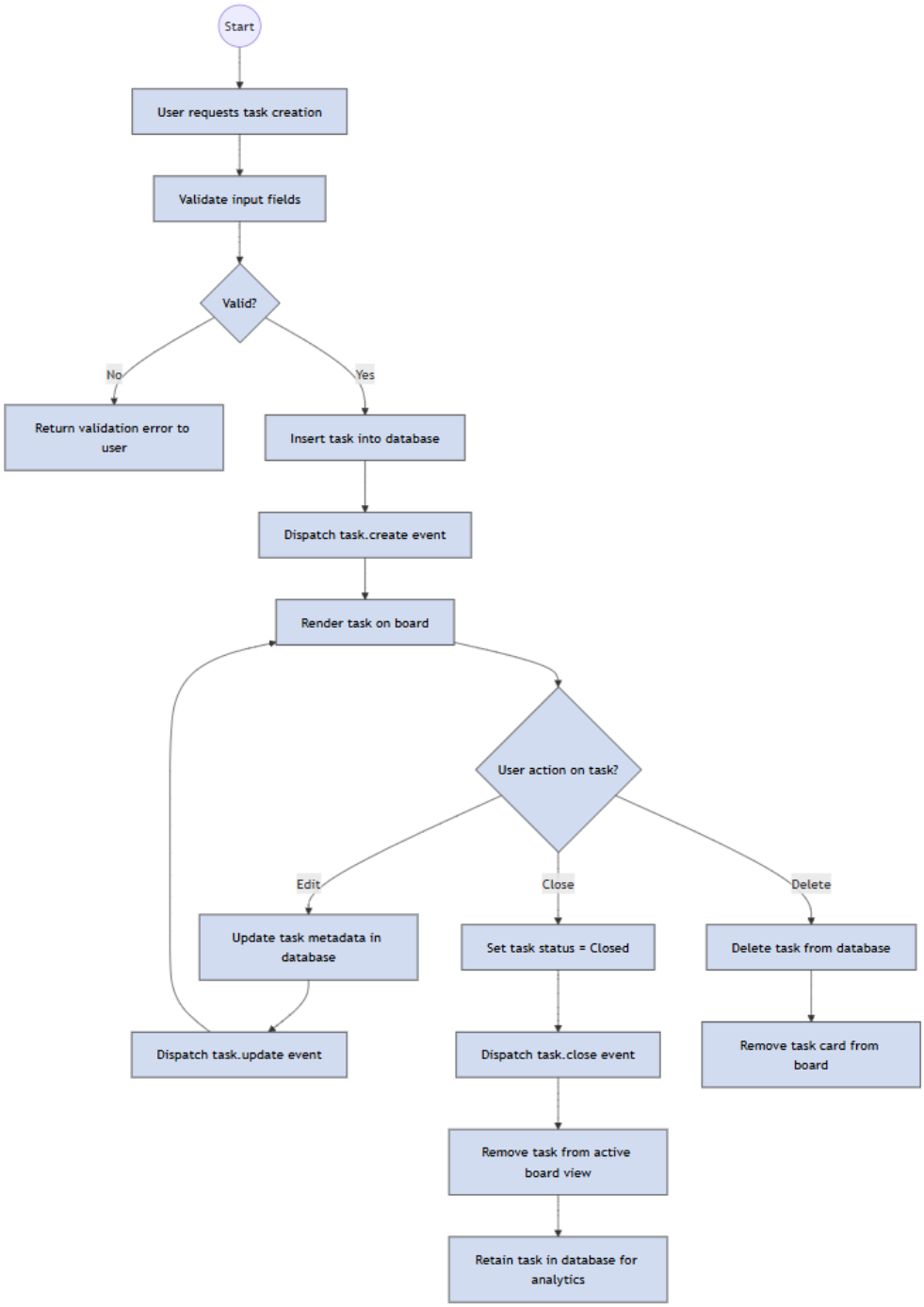
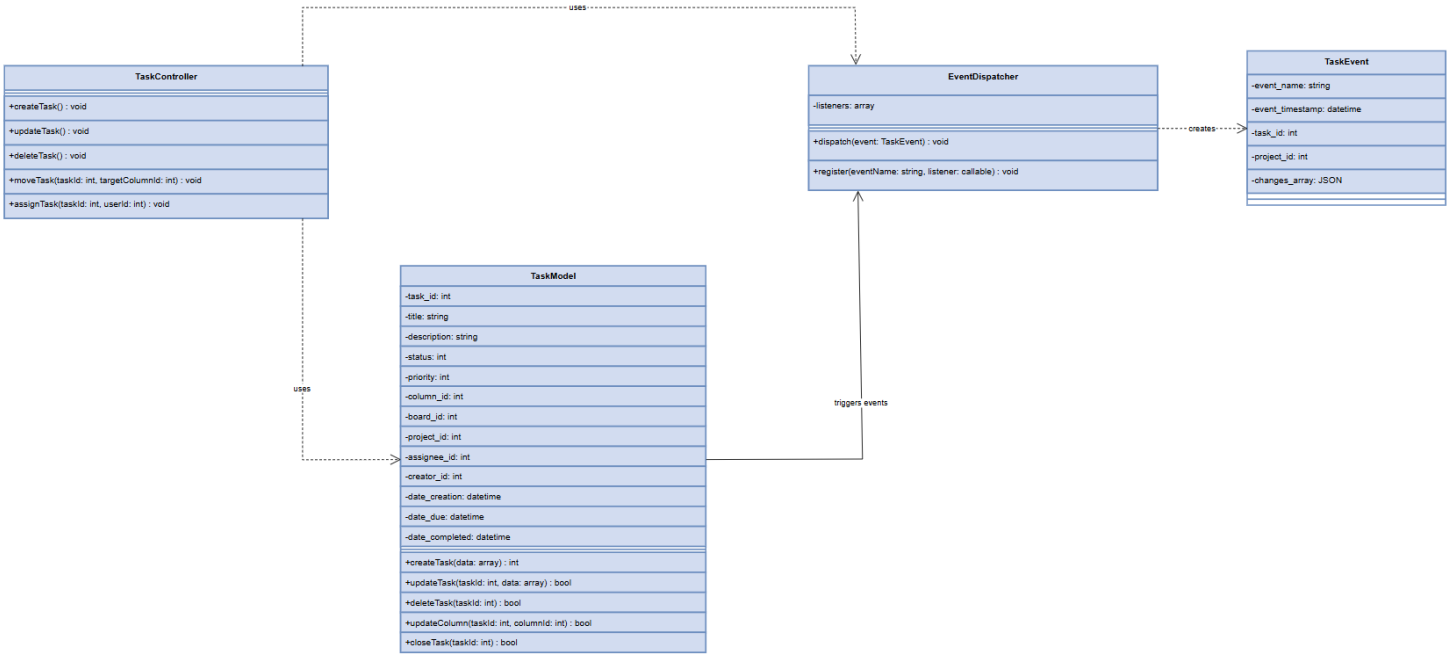


5.1.4.3 Activity Diagram

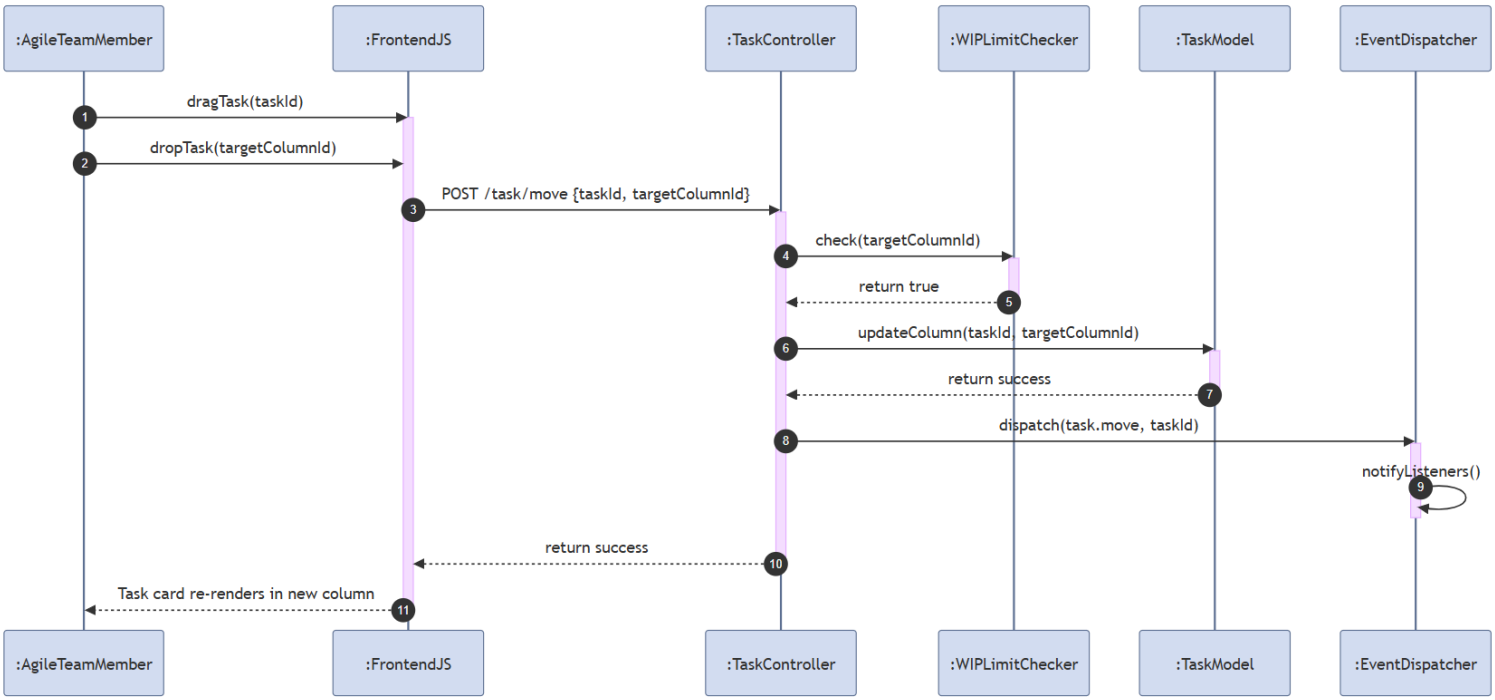


5.1.4.4 Class Diagram (module-level detailed class view)

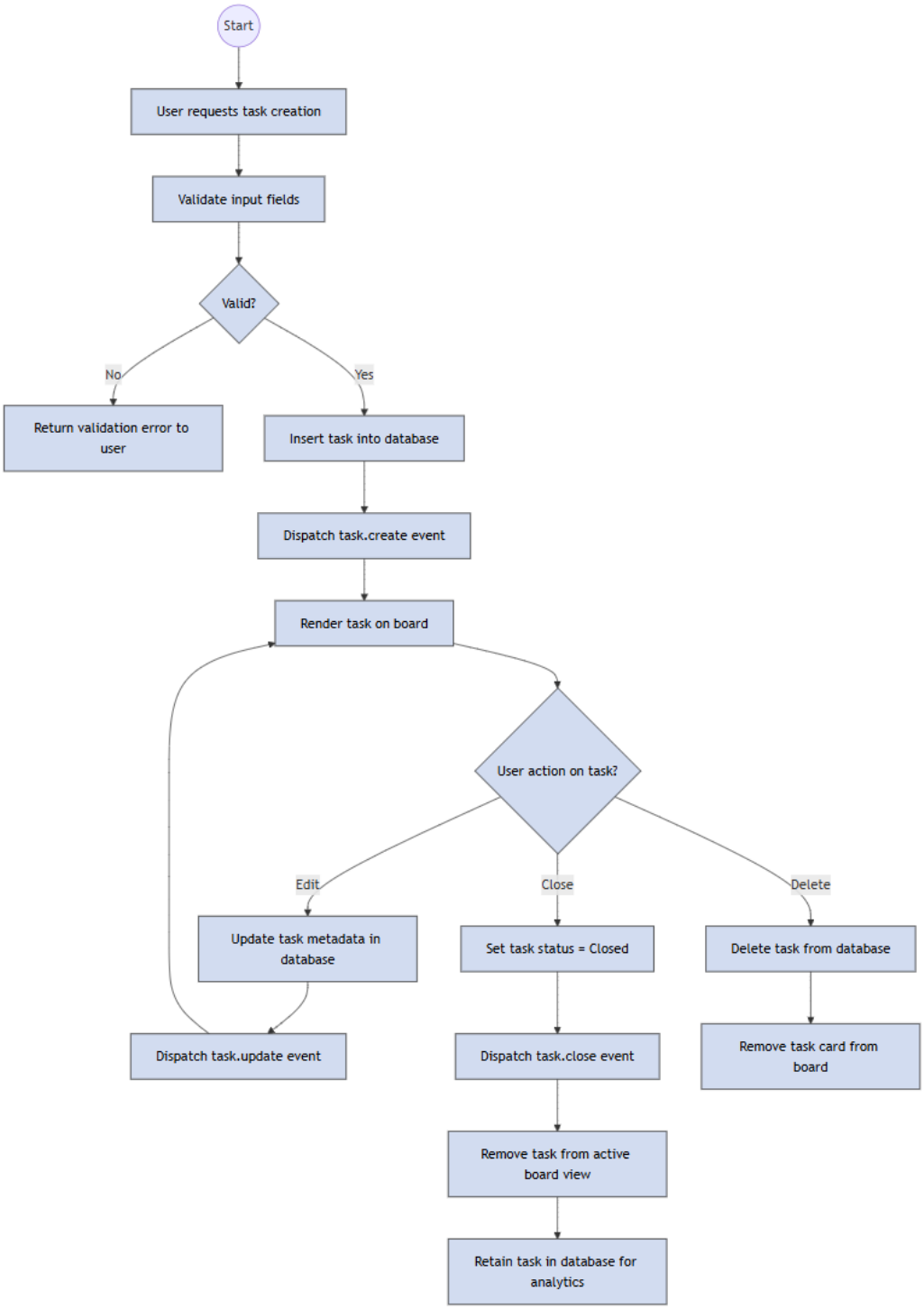


5.1.5 Task Assignment and Workflow Module Detailed Design

5.1.5.1 Sequence Diagram



5.1.5.2 Collaboration (Communication) Diagram



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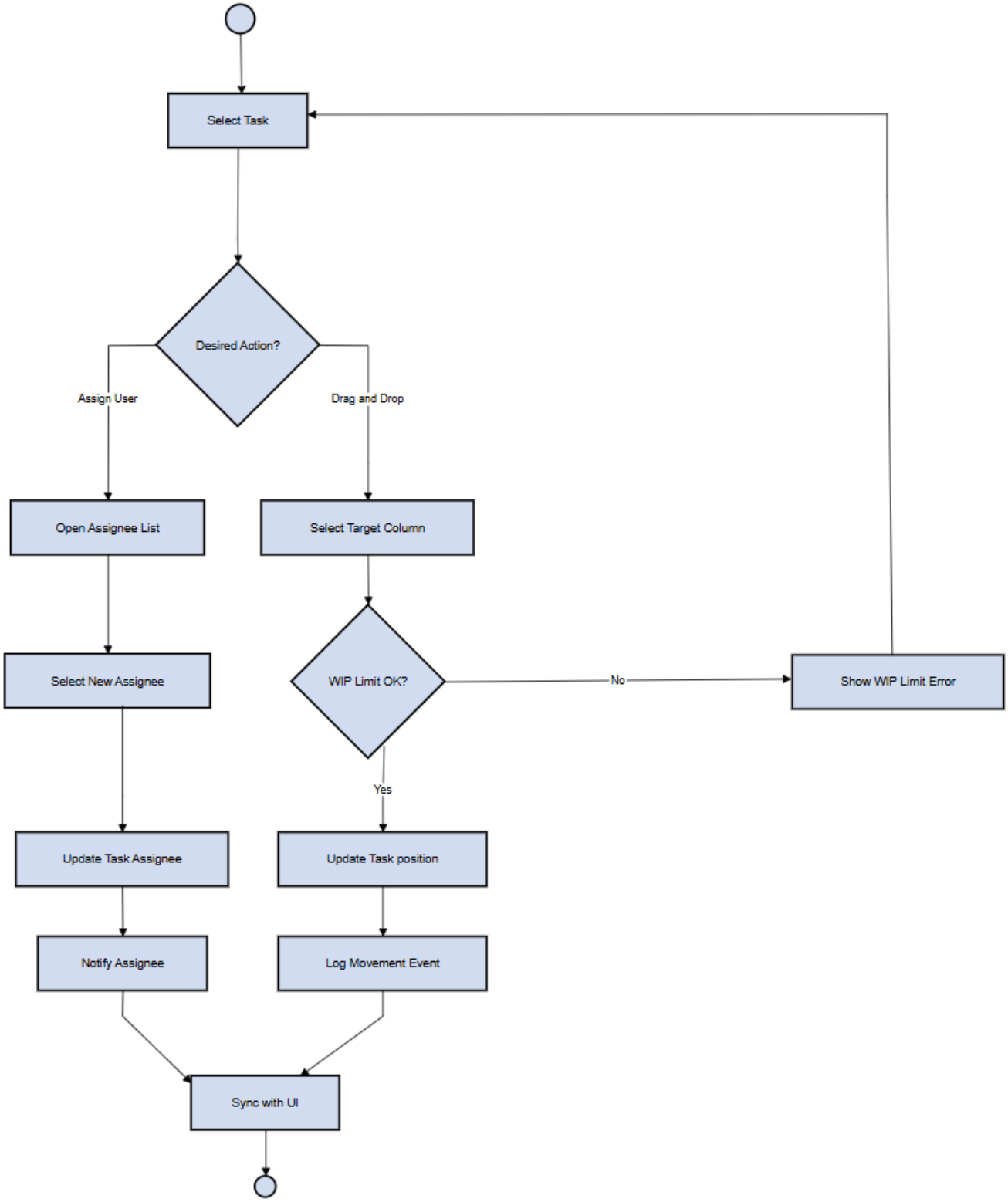
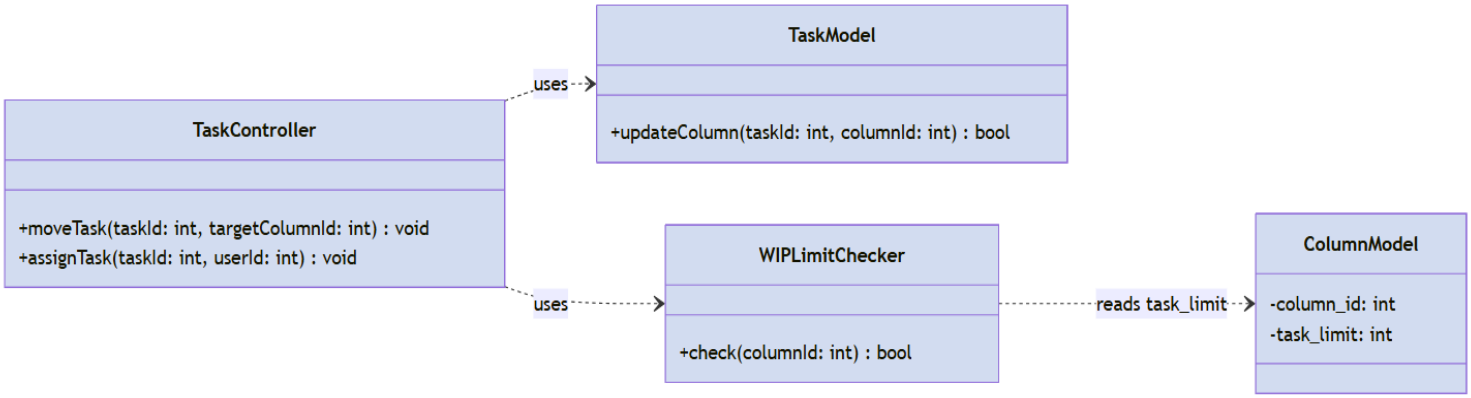
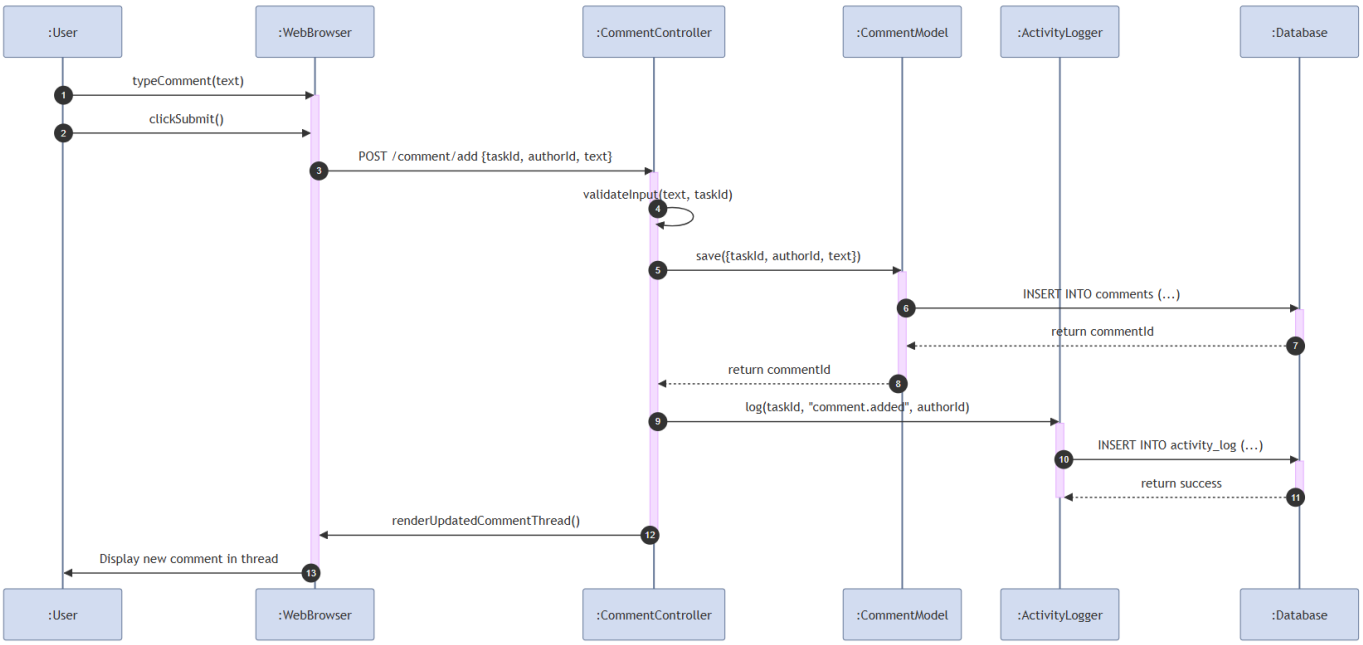


5.1.5.4 Class Diagram (module-level detailed class view)



5.1.6 Comment and Collaboration Module Detailed Design

5.1.6.1 Sequence Diagram



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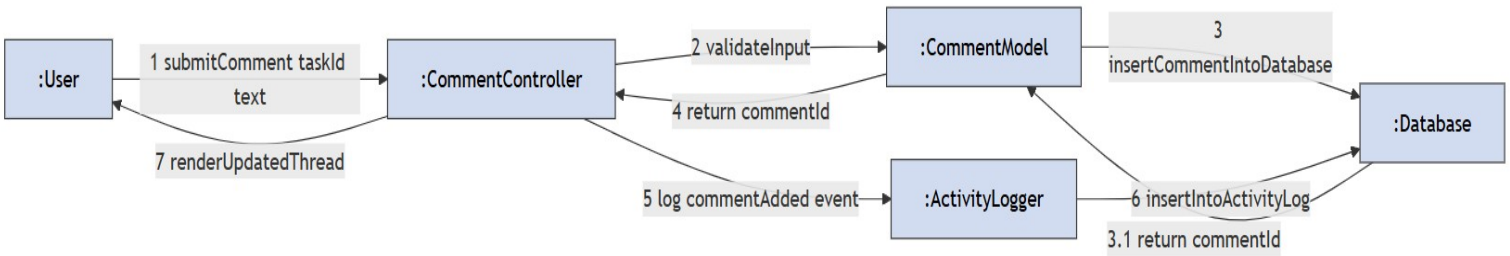
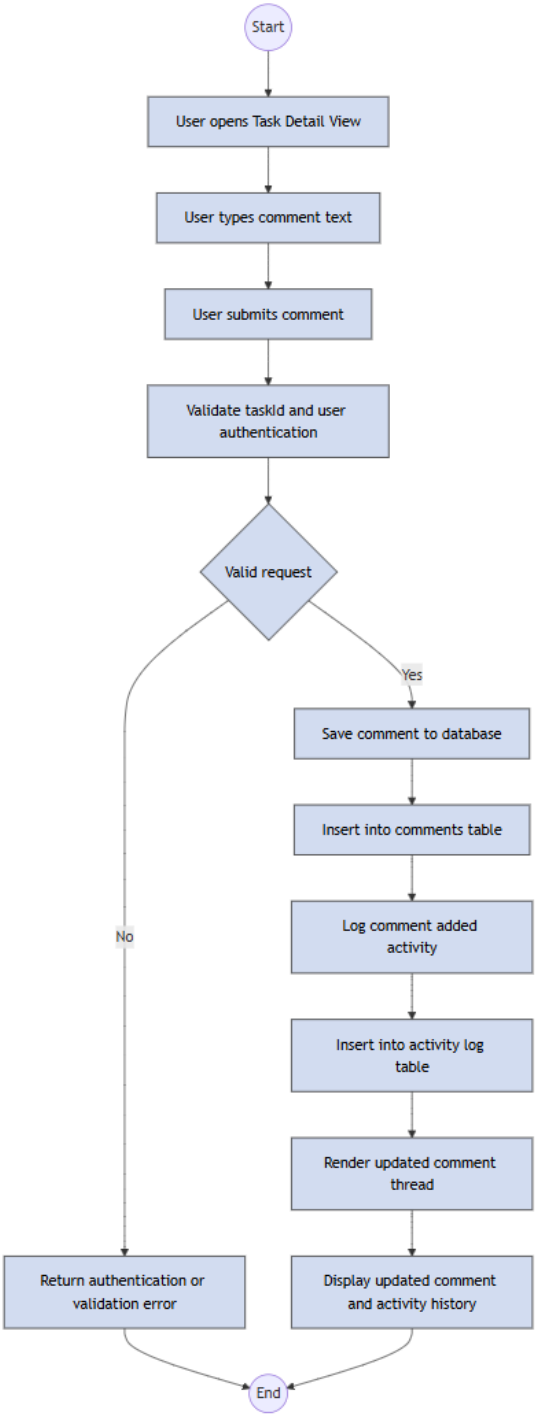
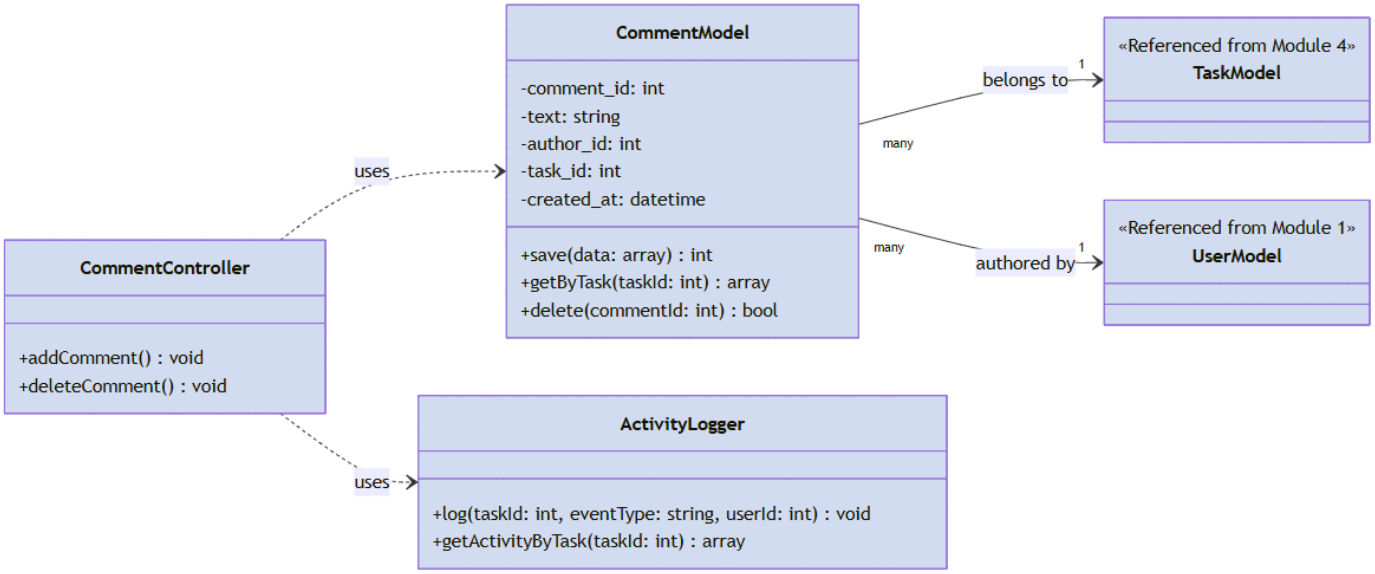


5.1.6.3 Activity Diagram

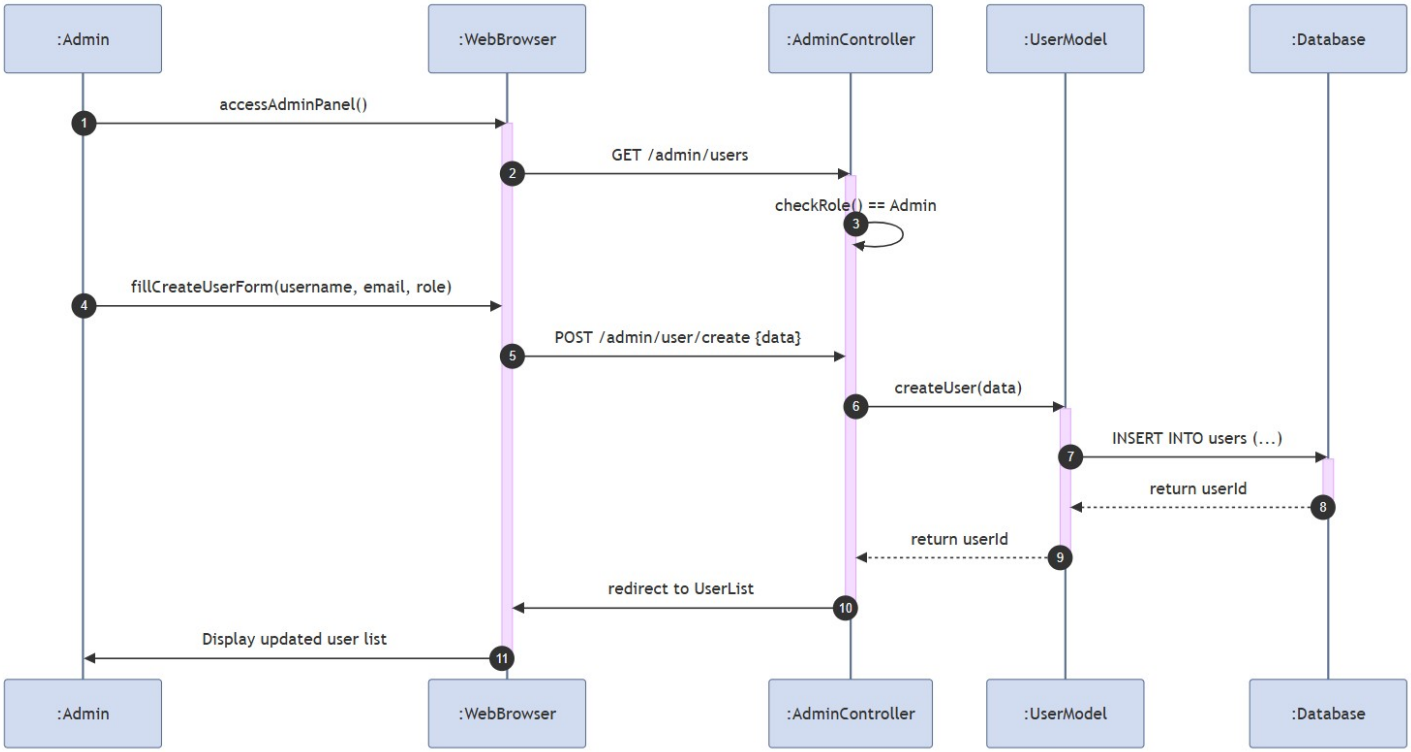


5.1.6.4 Class Diagram (module-level detailed class view)

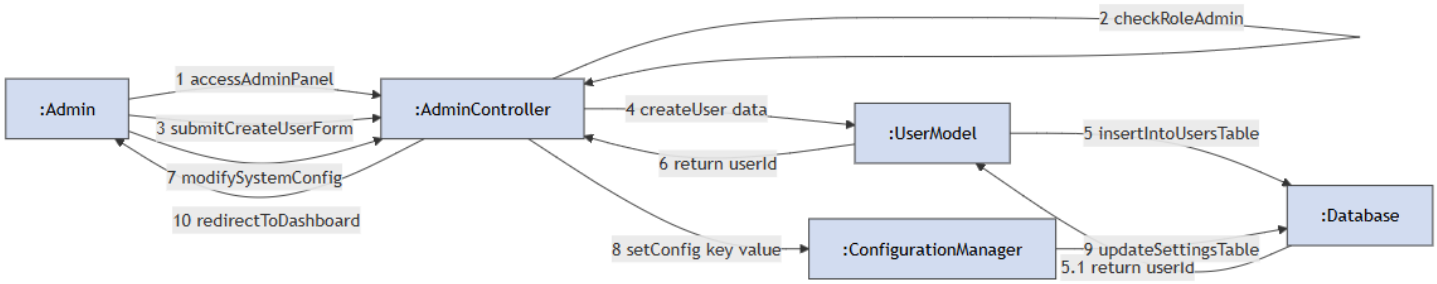


5.1.7 Admin Management Module Detailed Design

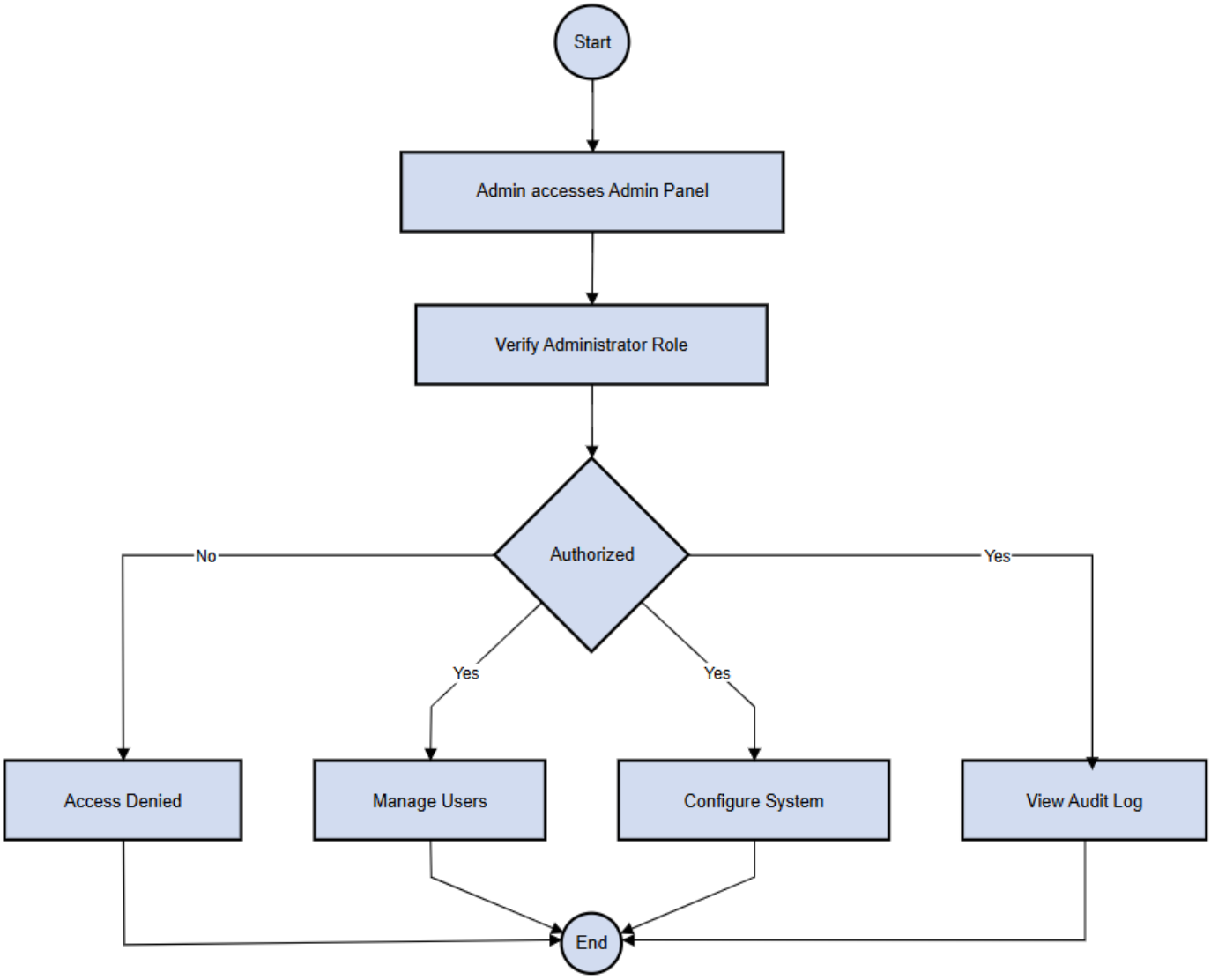
5.1.7.1 Sequence Diagram



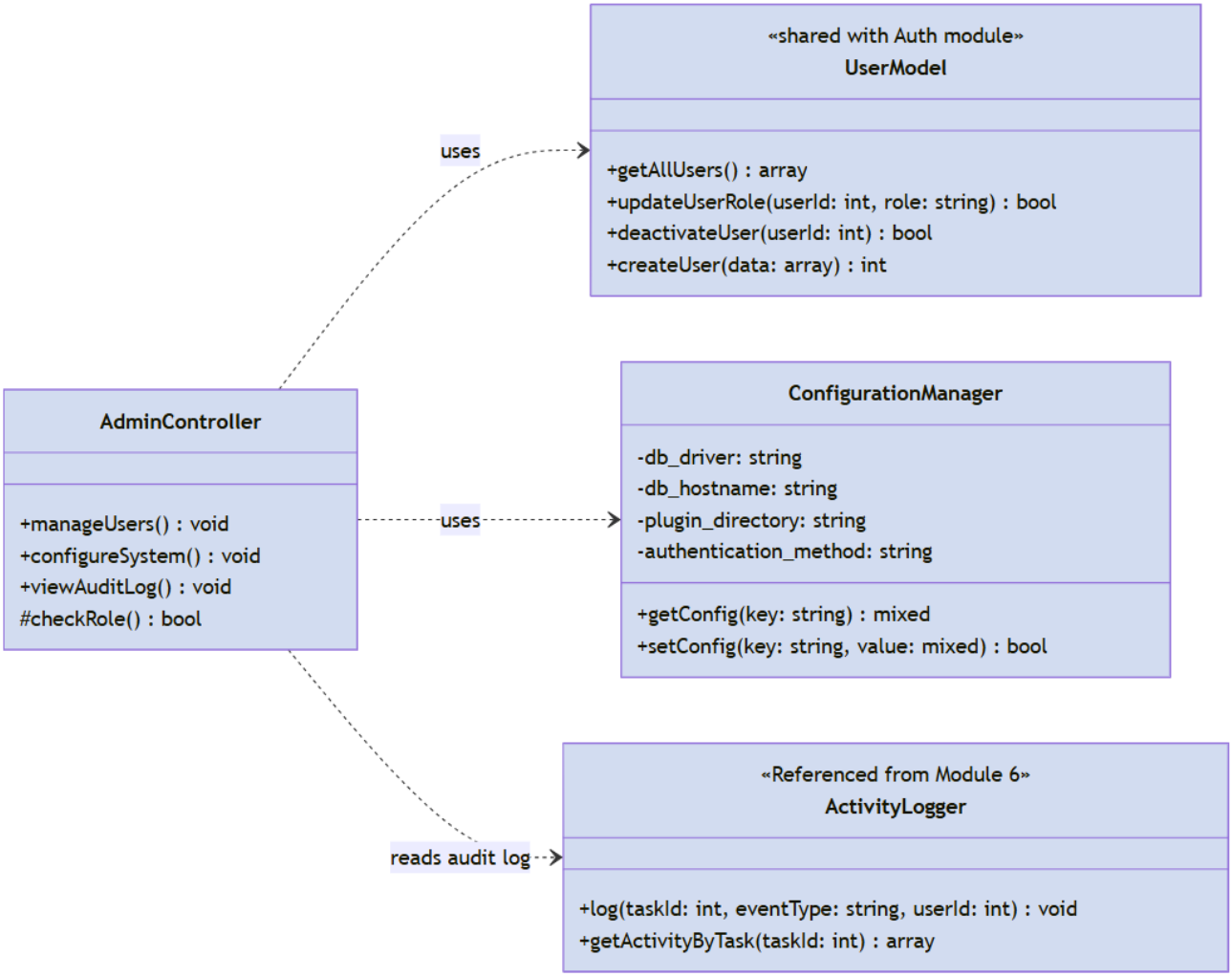
5.1.7.2 Collaboration (Communication) Diagram



5.1.7.3 Activity Diagram

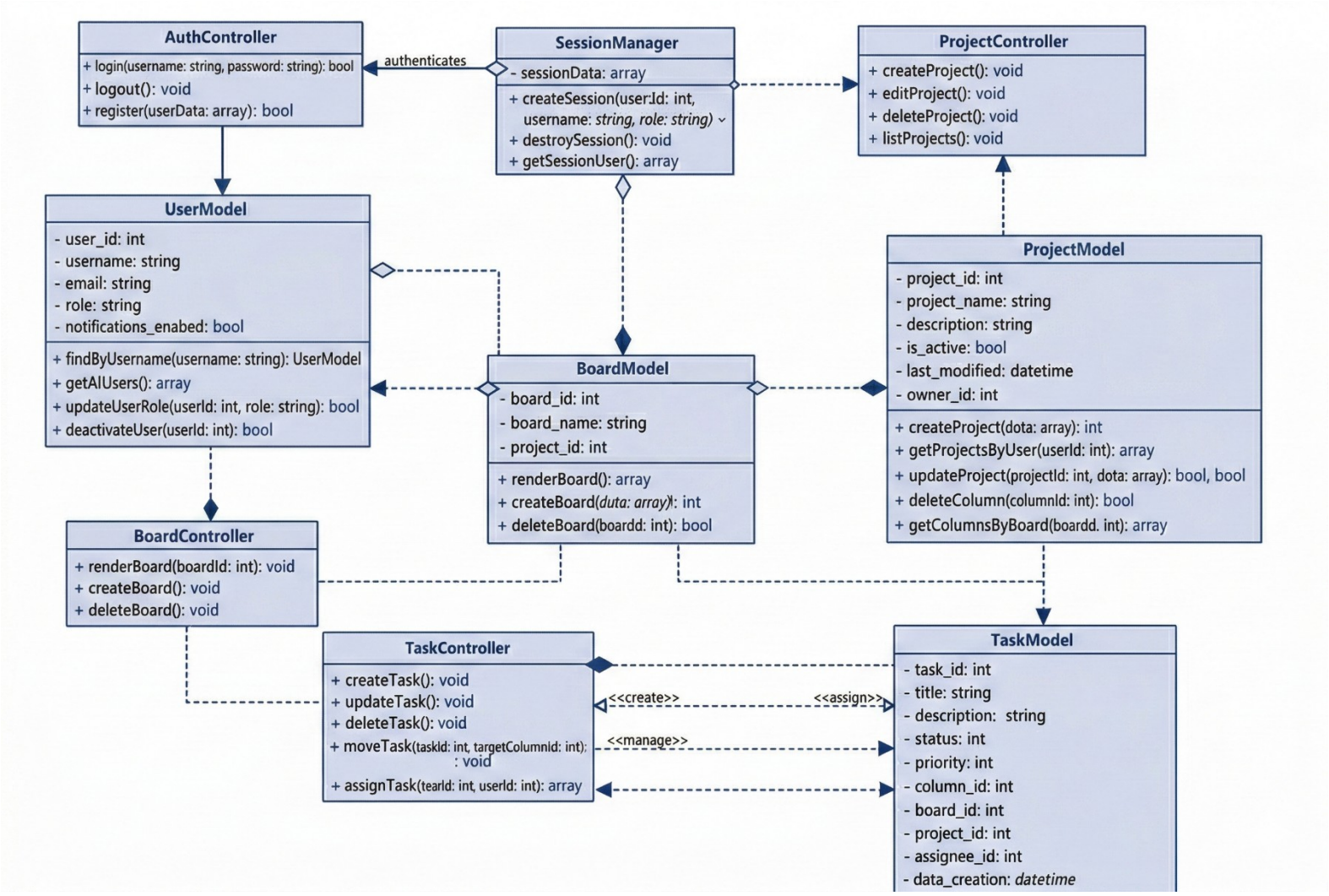


5.1.7.4 Class Diagram (module-level detailed class view)

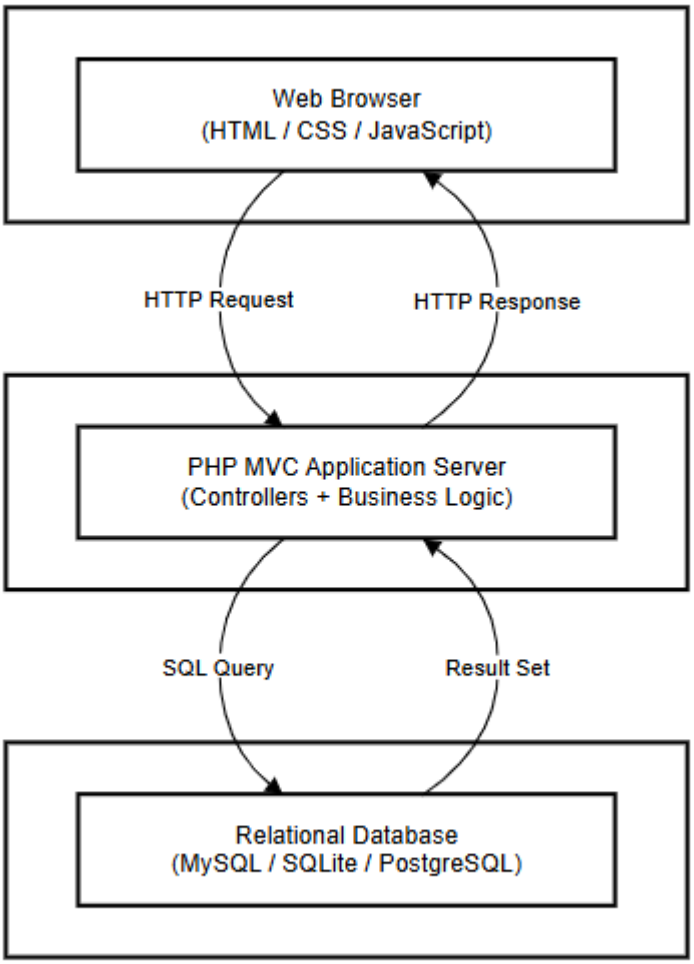


Appendix A — Kanboard System Class Diagram

The following full system class diagram illustrates the relationships between all primary classes in the Kanboard Project Management System. Refer to the UML Diagram Design Reference document for Modelio drawing instructions.



Appendix B — System Architecture Overview

The Kanboard Project Management System follows a three-tier architectural model:

- Presentation Tier: HTML/CSS/JavaScript rendered in the user's web browser. Handles UI events and dispatches HTTP requests to the application tier.
- Application Tier: PHP MVC application server. Controllers receive requests, invoke Models, and return rendered Views. Implements all business logic and RBAC enforcement.
- Data Tier: Relational database (MySQL/SQLite/PostgreSQL). Stores all persistent system data. Accessed exclusively through Model classes using parameterised queries.